

Qingyun Yang

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Undergraduate researcher addressing **clinically meaningful** problems through **computer vision**, **graphics**, and **machine learning**.

EDUCATION

Vanderbilt University **B.S.**
Majors: Computer Science, Mathematics | *Minor:* Data Science
GPA: 3.94/4.0 (Dean’s List All Semesters)

Aug 2022 - Present (expected May 2026)
Nashville, Tennessee

RESEARCH EXPERIENCE

Physics-Informed Predictive Modeling of Brain Tumor Growth from Single MRI Scans
Advisor: Prof. Peirong Liu, Johns Hopkins University
Student Researcher | **Predictive Modeling, Physics-Informed Learning**

April 2025 – Present

- Designed a framework using Physics-Informed Neural Networks and learning-based deformable registration to extract brain tumor dynamics from cross-subject single-timepoint MRI, enabling tumor progression prediction when longitudinal data is unavailable.

3D Deformation Modeling and Registration of Tissue Specimens for Surgical Guidance
Advisor: Prof. Jie Ying Wu, Vanderbilt University
Student Researcher | **Computer Vision, Graphics, Computer-assisted Interventions**

January 2024 – Present

- Led an independent research project on biomechanic-based deformable registration to correct tissue deformation for surgical visual guidance, improving alignment by **54%** over rigid registration.
- Established a novel deformable registration framework that uses both pre- and post-resection site surface point clouds as targets, boosting alignment accuracy by **33%** over previous deformable registration method in clinically challenging tongue cases.
- Developed a promptable 3D point cloud segmentation and reconstruction script from stereo images using the Segment Anything Model, streamlining workflow and improving segmentation efficiency.
- Implemented algorithm using PCA projection to automatically extract contour polylines from 3D point clouds for correspondence-based deformable registration.
- Built a rigid registration algorithm combining Iterative Closest Point (ICP) and fiducial-based registration, enabling robust alignment of small specimens with a limited number of fiducial points.
- Conducted **11** cadaver studies, collecting **33** En Bloc tissue specimens and reconstructing corrected 3D specimen models.

Generalizable Deep Regression Model Across Multi-Contrast CT for Body Part Localization
Advisor: Prof. Bennett A. Landman, Vanderbilt University
Research Assistant | **Machine Learning, Medical Imaging**

August 2024 – January 2025

- Investigated the generalizability of deep regression models across multi-contrast CT by designing **3** experimental groups with variations of Global-Intensity-Nonlinear (GIN) augmentation, showing that GIN improves cross-phase model performance.
- Conducted rigorous assessment using RMSE calculations and paired t-tests, showing that GIN in pre-training phase alone yields comparable generalization to applying GIN in both phases, highlighting efficient augmentation strategies.

PUBLICATIONS & PRESENTATIONS

- Yang, Q.***, Li, F.* , Xu, J., Liu, Z., Sridhar, S., Jin, W., ... & Wu, J. Y. (2025, September). Augmented Reality-Based Guidance with Deformable Registration in Head and Neck Tumor Resection. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 45-54). Cham: Springer Nature Switzerland. **MICCAI. (accepted paper, oral presentation, paper highlight)**
- Yang, Q.**, Acar, A., Ringel, M. J., Heiselman, J. S., Miga, M. I., Topf, M., & Wu, J. Y. (2025, April). Nonrigid alignment of en bloc tissue specimen to resection bed to enhance correspondence for re-resection guidance. In *Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling* (Vol. 13408, pp. 238-246). **SPIE. (accepted paper, oral presentation)**
- Li, H., Wang, J., Kumar, N., d’Almeida, J., Lu, D., Acar, A., Han, J., **Yang, Q.**, ... & Oguz, I. (2025, April). Automated segmentation of central airway obstruction from endoscopic video stream with deep learning. In *Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling* (Vol. 13408, pp. 113-119). **SPIE.**
- Su, D., **Yang, Q.**, Van Schaik, K. D., Remedios, L. W., Zhang, Y., Li, T., Maldonado, F., Sandler, K. L., Dawant, B. M., & Landman, B. A. (n.d.). BPPR: A framework for content navigation in multi-contrast body CT images using deep regression models. **IEEE Transactions on Biomedical Engineering.** (manuscript under review)

HONORS & AWARDS

CRA Outstanding Undergraduate Researcher Award

- **2026 Awardee** – Selected as one of 8 undergraduates from hundreds of nominees across North American universities for exceptional computing research
- **2025 Honorable Mention**

MICCAI 2025 NIH Registration Grant

- Competitively selected by NIH/NIBIB principal investigators in coordination with MICCAI Society and conference organizers, supporting participation in the MICCAI 2025 conference.

Stanford Exposure to Research and Graduate Education (SERGE) Scholar

- Chosen for a highly selective Stanford Engineering program providing a fully funded PhD preparation experience.

Vanderbilt Institute for Surgery and Engineering (VISE) Summer Fellowship

- Competitively selected twice (**2024 & 2025**), receiving **\$8,000** each year to support independent research.

1st Place, Video & Poster Presentation Award

- Awarded first place among undergraduate research posters at VISE Annual Surgery, Intervention, and Engineering Symposium.

WiC's Grace Hopper Celebration Scholarship

- Received full funding from Vanderbilt Women in Computing to attend Grace Hopper Celebration conference, providing professional development at the world's largest conference for women in computing.

TEACHING EXPERIENCE

Vanderbilt Computer Science Department

Teaching Assistant / Course: Programming and Problem Solving (Java) August 2023 – Present

- Conducted **4** hours of weekly office hours, addressing inquiries from over **200** students across **3** sessions, enhancing students' understanding of programming concepts.
- Evaluated assignments and exams, offering detailed feedback on syntax, logic, and coding style, ensuring students develop strong programming skills and promoting best practices in code quality.

Vanderbilt Student Volunteers for Science

Volunteer Teacher February 2023 – April 2023

- Taught weekly 70-minute science lessons to 20+ middle school students, incorporating hands-on experiments to reinforce scientific concepts and critical thinking skills.
- Adapted lesson plans to diverse learning styles, fostering student engagement and curiosity in STEM fields.

TECHNICAL SKILLS

Languages: Python, C++, Java, R, Racket, Prolog, MATLAB

Technical Skills: deformable registration (biomechanics-based, learning-based), rigid registration (ICP, fiducial-based), point cloud processing, image segmentation, domain generalization, deep regression models, disentangled representation learning